Advanced Risk Minimization

Regression Losses: L2 and L1 loss

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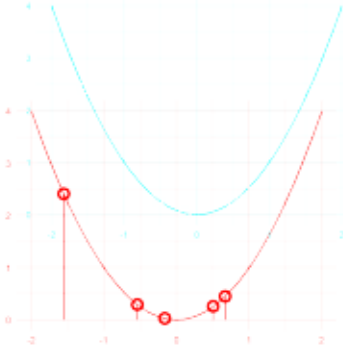

Learning goals

- Derive the risk minimizer of the L2-loss
- Derive the optimal constant model for the L2-loss
- Know risk minimizer and optimal constant model for L1-loss

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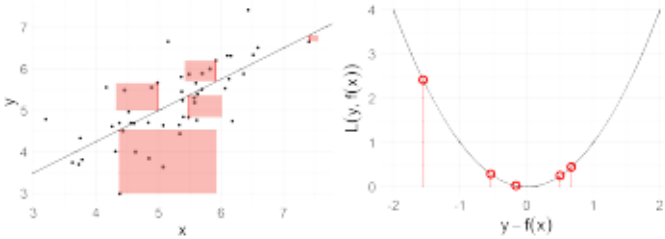
L2-LOSS

$$L(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2 \quad \text{or} \quad L(y, f(\mathbf{x})) = 0.5\,(y - f(\mathbf{x}))^2$$

- Tries to reduce large residuals (if residual is twice as large, loss is 4 times as large), hence outliers in y can become problematic
- Analytic properties: convex, differentiable $\Rightarrow$ gradient no problem in loss minimization
  (**Warning**: $\mathcal{R}_{\text{emp}}(f)$ can still be non-smooth/non-convex due to $f(\mathbf{x})$)



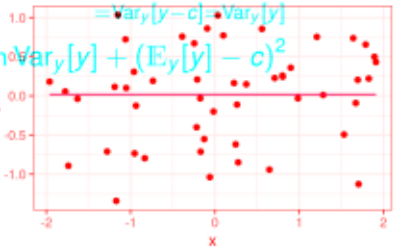
L2-LOSS: OPTIMAL CONSTANT MODEL

Let us consider the (true) risk for $\mathcal{Y} = \mathbb{R}$ and $L2$-Loss $L(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2$ with $\mathcal{H}$ restricted to constants. As derived previously, for the squared loss, the optimal constant model is given by

$$f_c^* = \arg\min_{c \in \mathbb{R}} \mathbb{E}_{xy}\left[(y - c)^2\right] = \mathbb{E}_y[y]$$

We also derived: the empirical constant minimizer is $\hat{f}_c = \bar{y}$.

$$\begin{aligned} f_c^* &= \arg\min_{c \in \mathbb{R}} \mathbb{E}_{xy}\left[(y - c)^2\right] = \arg\min_{c \in \mathbb{R}} \mathbb{E}_y\left[(y - c)^2\right] \\ &= \arg\min_{c \in \mathbb{R}} \underbrace{\mathbb{E}_y\left[(y - c)^2\right]}_{\text{loss}} = (\mathbb{E}_y[y] - c)^2 + (\mathbb{E}_y[y] - c)^2 \\ &= \arg\min_{c \in \mathbb{R}} \mathrm{Var}_y[y] + (\mathbb{E}_y[y] - c)^2 \\ &= \mathbb{E}_y[y] \end{aligned}$$



L2-LOSS: RISK MINIMIZER

Let us consider the (true) risk for $\mathcal{Y} = \mathbb{R}$ and the $L2$-Loss $L(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2$ with unrestricted $\mathcal{H} = \{f(\mathbf{x}) : \mathcal{X} \to \mathbb{R}^g\}$.

Since $\mathcal{H}$ is unrestricted, at any point $\mathbf{x} = \tilde{\mathbf{x}}$, we can predict any value c we want. To find

$$f^*(\tilde{\mathbf{x}}) = \text{argmin}_c \mathbb{E}_{y|x}\left[L(y, c) \mid \mathbf{x} = \tilde{\mathbf{x}}\right],$$

we can apply the previous derivation of f_c^* under $L2$ loss using the conditional label distribution. The best point-wise prediction is then given by the cond. mean

$$f^*(\tilde{\mathbf{x}}) = \mathbb{E}_{y|x}\left[y \mid \mathbf{x} = \tilde{\mathbf{x}}\right].$$

L2-LOSS MEANS MINIMIZING VARIANCE

Rethinking what we did in the opt. constant model: We optimized for a constant whose squared distance to all data points is minimal (in sum, or on average). This turned out to be the mean.

For the optimal constant model f_c for the L2-loss $L(y, f) = (y - f)^2$ we solve the optimization problem

$$\arg\min_{\theta \in \mathbb{R}} \mathcal{R}_{\text{emp}}(f) = \arg\min \sum_{i=1}^{n} (y^{(i)} - \theta)^2.$$

What if we calculcate the loss of $\hat{\theta} = \bar{y}$? That's $\mathcal{R}_{\text{emp}} = \sum_{i=1}^{n} (y^{(i)} - \bar{y})^2$.

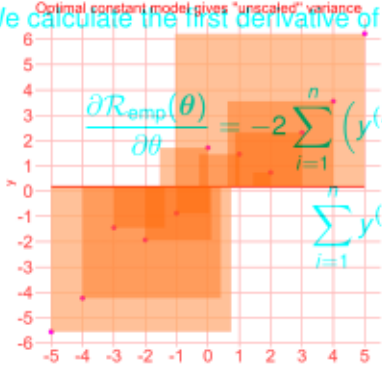
Average this by $\frac{1}{n}$ or $\frac{1}{n-1}$ to obtain variance.

We calculate the first derivative of $\mathcal{R}_{\text{emp}}$ w.r.t. θ and set it to 0:

$$\frac{\partial \mathcal{R}_{\text{emp}}(\theta)}{\partial \theta} = -2 \sum_{i=1}^{n} \left(y^{(i)} - \theta\right) = 0$$

$$\sum_{i=1}^{n} y^{(i)} - n\theta = 0$$

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^{n} y^{(i)} =: \bar{y}.$$



- Generally, if model yields unbiased predictions, $\mathbb{E}_{y \mid \mathbf{x}}[y - f(\mathbf{x}) \mid \mathbf{x}] = 0$, using $L2$-loss means minimizing variance of model residuals
- Same holds for the pointwise construction / conditional distribution considered before

L1-LOSS: RISK MINIMIZER

The L1 loss is defined as

$$L(y, f(\mathbf{x})) = |y - f(\mathbf{x})|$$

- More robust than $L2$, outliers in y are less problematic.
- Analytical properties: convex, not differentiable for $y = f(\boldsymbol{x})$ (optimization becomes harder).

Let us consider the (true) risk for $\mathcal{Y} = \mathbb{R}$ and the $L2$-Loss $L(y, f) = (y - f)^2$ with unrestricted $\mathcal{H} = \{f : \mathcal{X} \to \mathbb{R}^g\}$.

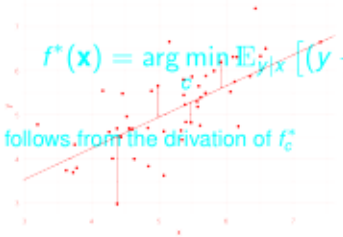
- By the law of total expectation

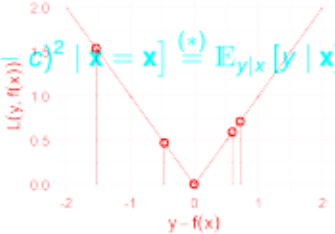
$$\mathcal{R}_L(f) = \mathbb{E}_{xy}\left[L(y, f(\mathbf{x}))\right] = \mathbb{E}_x\left[\mathbb{E}_{y|x}\left[L(y, f(\mathbf{x})) \mid \mathbf{x} = \mathbf{x}\right]\right]$$

$$= \mathbb{E}_x\left[\mathbb{E}_{y|x}\left[(y - f(\mathbf{x}))^2 \mid \mathbf{x} = \mathbf{x}\right]\right]$$

- Since $\mathcal{H}$ is unrestricted, at any point $\mathbf{x} = \mathbf{x}$, we can predict any value c we want. The best point-wise prediction is the cond. mean

$$f^*(\mathbf{x}) = \arg\min_c \mathbb{E}_{y|x}\left[(y - c)^2 \mid \mathbf{x} = \mathbf{x}\right] \overset{(*)}{=} \mathbb{E}_{y|x}\left[y \mid \mathbf{x}\right].$$





$^{(*)}$ follows from the drivation of f_c^*

L1-LOSS: RISK MINIMIZER

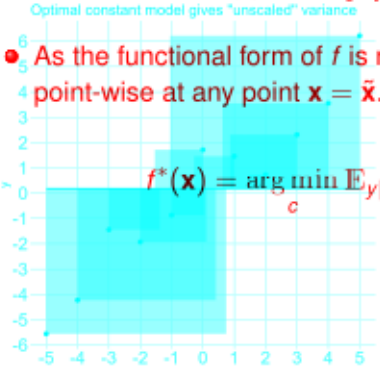
We calculate the (true) risk for the $L1$-Loss $L(y, f(\mathbf{x})) = |y - f(\mathbf{x})|$ with unrestricted $\mathcal{H} = \{f : \mathcal{X} \to \mathbb{R}\}$.

- We use the law of total expectation

$$\mathcal{R}(f) = \mathbb{E}_x \left[\mathbb{E}_{y|x} \left[|y - f(\mathbf{x})| \; | \mathbf{x} = \tilde{\mathbf{x}} \right] \right].$$

- As the functional form of f is not restricted, we can just optimize point-wise at any point $\mathbf{x} = \tilde{\mathbf{x}}$. The best prediction at $\mathbf{x} = \tilde{\mathbf{x}}$ is then

$$f^*(\mathbf{x}) = \arg\min_c \mathbb{E}_{y|x} \left[|y - c| \right] = \text{med}_{y|x} \left[y \mid \mathbf{x} \right].$$



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L1-LOSS: OPTIMAL CONSTANT MODEL

The L1 loss is defined as

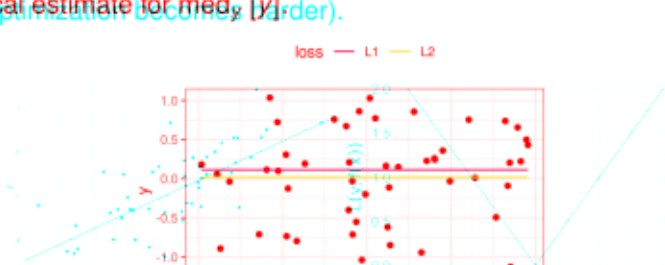
$$L(y, f) = |y - f|$$

The optimal constant model in terms of the theoretical risk for the L1 loss is the median over y:

$$f_c^* = \text{med}_{y|x}[y \mid \mathbf{x}] \overset{\text{drop } \mathbf{x}}{=} \text{med}_y[y]$$

- More robust than L2, outliers in y are less problematic.
- Analytical properties: convex, not differentiable for $y = f(x)$ (optimization becomes harder).

The optimizer $\hat{f}_c$ of the empirical risk is $\text{med}(y^{(i)})$ over $y^{(i)}$, which is the empirical estimate for $\text{med}_y[y]$.



L1-LOSS: RISK MINIMIZER

We calculate the (true) risk for the $L1$-Loss $L(y, f) = |y - f|$ with unrestricted $\mathcal{H} = \{f : \mathcal{X} \to \mathcal{Y}\}$.

- We use the law of total expectation

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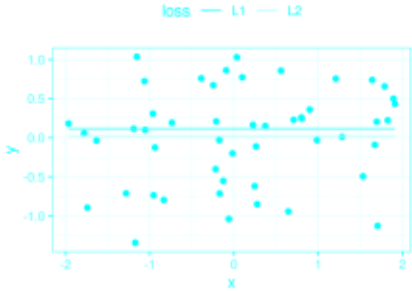
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